COVID Mortality Analysis Project

Dane Pearson, Dhriti Avala

2025-12-02

Statistical Analysis of US COVID-19 Mortality Rates in R:

Install packages:

```
library(lubridate)
library(ggplot2)
library(ggpubr)
library(tidyr)
library(maps)
library(patchwork)
library(dplyr)

# For formatting tables nicely
library(knitr)
library(flextable)
library(modelsummary)
library(kableExtra)
```

Load data:

```
original_df <- read.csv(
  "data/COVID-19_Deaths_by_Sex_Age.csv",
  stringsAsFactors = TRUE
)
```

```r
state_populations_data <- read.csv(
  "data/state_populations.csv",
  stringsAsFactors = TRUE
)

# Civilian population by state, single year of age, and sex (Census Vintage 2025).
# Lets each age-sex group's deaths be divided by THAT group's population.
age_sex_pops_df <- read.csv("data/age_sex_state_pops.csv")
```

EDA:

Make summary tables of both datasets:

```r
original_summary <- summary(original_df)
state_pop_summary <- summary(age_sex_pops_df)

kable(
  original_summary,
  caption = "Original dataset summary",
  digits = 1,
  align = "l"
)
```

Table 1: Original dataset summary

Data As Of	Start Date	End Date	Group	Year	Month	State	Sex	Age Group	COVID-19 Deaths	Total Deaths	Pneumonia Deaths	Pneumonia and COVID-19 Deaths	Influenza Deaths	Pneumonia, Influenza, or COVID-19 Deaths	Footnote
09/27/2026: 8262	01/01/2020: 8262	09/26/2026: 8262	Monthly: 8262	Min. :2020	1 : 1130	Alabama: 1130	All Sexes: 1130	0-17 years: 1130	Min. : 0	Min. : 0	Min. : 0	Min. : 0	Min. : 0	Min. : 0	COVID-19 Deaths and Influenza Deaths: 1
				1st Qu.:2022	3 : 1130	Alabama-Autauga County: 1	Female: 1130	18-29 years: 1130	1st Qu.: 0	1st Qu.: 322	1st Qu.: 0	1st Qu.: 0	1st Qu.: 0	1st Qu.: 509	Pneumonia and COVID-19 Deaths and Influenza Deaths: 1
				Median :2023	5 : 1130	Alabama-Baldwin County: 1	Male: 1130	30-39 years: 1130	Median : 0	Median : 930	Median : 0	Median : 0	Median : 0	Median : 1	COVID-19 Deaths and Pneumonia Deaths: 1
				Mean :2023	7 : 1130	Alabama-Barbour County: 1		40-49 years: 1130	Mean : 50	Mean : 2509	Mean : 44	Mean : 6	Mean : 0	Mean : 103	COVID-19 Deaths and Pneumonia and Influenza Deaths: 1
				3rd Qu.:2024	9 : 1130	Alabama-Blount County: 1		50-64 years: 1130	3rd Qu.: 0	3rd Qu.: 1759	3rd Qu.: 0	3rd Qu.: 0	3rd Qu.: 0	3rd Qu.: 435	Pneumonia and Influenza Deaths: 1
				Max. :2026	11 : 1130	Alabama-Bullock County: 1		65-74 years: 1130	Max. :6503	Max. :22509	Max. :650	Max. :193	Max. :19	Max. :1034	NA
					NA	Alabama-Butler County: 1		75-84 years: 1130							NA
					NA	Alabama-Calhoun County: 1		85 years and over: 1130							NA

Data.As.Of	Start.Date	End.Date	Group	Year	Month	State	Sex	Age.Group	COVID.19.Deaths	Total.Deaths	Pneumonia.Deaths	Pneumonia.and.COVID.19.Deaths	Influenza.Deaths	Pneumonia..Influenza..or.COVID.19.Deaths	Footnote
NA	01/01/2020: 5508	12/31/2020: 5508	Total: 2754	2020: 3.0	1st Qu.:2020	Alaska: 2550	Female: 45900	1-4 years: 8100	:39430	0 :2858	0 :36201	:36884	0 :26688	0 :30828	One or more data cells have counts between 1-9 and have been suppressed in accordance with NCHS confidentiality standards.:97896
NA	01/01/2020: 5508	12/31/2020: 5508	Year: 11016	Median :2021	Median 6.0	Arizona: 2550	Male: 45900	15-24 years: 8100	10 :1491	10 :1141	10 :816	10 : 1347	10 :65	10 : 1660	NA
NA	01/01/2020: 5508	12/31/2020: 5508	NA	Mean :2021	Mean 6.2	Arkansas: 2550	NA	18-29 years: 8100	11 :1332	11 :1332	11 :1431	11 : 1169	11 :381	11 : 1478	NA
NA	02/01/2020: 2754	12/31/2020: 2754	NA	3rd Qu.:2021	3rd Qu.: 9.0	California: 2550	NA	25-34 years: 8100	12 :1166	13 :1247	12 :1337	12 : 1065	12 :343	12 : 1434	NA
NA	02/01/2020: 2754	12/31/2020: 2754	NA	Max. :2021	Max. 12.0	Colorado: 2550	NA	30-39 years: 8100	13 :1097	12 :1244	13 :1274	13 : 968	13 :318	13 : 1295	NA
NA	(Other) :107406	NA	NA's :2754	NA's :13770	(Other) :224000	NA	(Other) :89100	(Other) :46800	(Other) :52020	(Other) :38124	(Other) :46800	(Other) :36086	(Other) :56772		

```
kable(
  state_pop_summary,
  caption = "State populations dataset summary",
  digits = 1,
  align = "l"
)
```

Table 2: State populations dataset summary

SUMLEV	REGION	DIVISION	STATE	NAME	SEX	AGE	ESTBASE2020_CIV	POPESTIMATE2020_CIV	POPESTIMATE2021_CIV	POPESTIMATE2022_CIV	POPESTIMATE2023_CIV	POPESTIMATE2024_CIV	POPESTIMATE2025_CIV
Min. :10.00	Min. :0.000	Min. :0.000	Min. : 0.00	Length:135572	Min. :0	Min. : 0.00	Min. : 585	Min. : 587	Min. : 617	Min. : 632	Min. : 658	Min. : 706	Min. : 768
1st Qu.:40.00	1st Qu.:2.000	1st Qu.:3.000	1st Qu.:15.00	Class :character	1st Qu.:0	1st Qu.: 21.00	1st Qu.: 12056	1st Qu.: 12072	1st Qu.: 12194	1st Qu.: 12332	1st Qu.: 12496	1st Qu.: 12683	1st Qu.: 12830
Median :40.00	Median :3.000	Median :5.000	Median :28.50	Mode :character	Median :1	Median : 43.00	Median : 31291	Median : 31346	Median : 31522	Median : 31778	Median : 32147	Median : 32550	Median : 32792
Mean :39.42	Mean :2.615	Mean :5.019	Mean :28.40	NA	Mean :1	Mean : 53.49	Mean : 194754	Mean : 194791	Mean : 195090	Mean : 196233	Mean : 197870	Mean : 199790	Mean : 200835
3rd Qu.:40.00	3rd Qu.:3.000	3rd Qu.:7.000	3rd Qu.:41.25	NA	3rd Qu.:2	3rd Qu.: 65.00	3rd Qu.: 66438	3rd Qu.: 66413	3rd Qu.: 66604	3rd Qu.: 66913	3rd Qu.: 67261	3rd Qu.: 67774	3rd Qu.: 68144
Max. :40.00	Max. :4.000	Max. :9.000	Max. :56.00	NA	Max. :2	Max. :999.00	Max. :3303997	Max. :3304634	Max. :3309708	Max. :3329088	Max. :3356872	Max. :3389435	Max. :3407168

First convert Covid-19 Death column from factor to numeric, and remove commas in numbers:

```
original_df$COVID.19.Deaths <- as.numeric(gsub(
  ",",
  "",
  original_df$COVID.19.Deaths
))

# Ensure it was converted correctly
# class(original_df$COVID.19.Deaths)

# The CDC age labels use en dashes ("15–24 years"), not hyphens ("15-24 years").
# Replace them so every filter in this file, written with plain hyphens, matches.
# "\u2013" is the code for an en dash.
original_df$Age.Group <- gsub(
  "\u2013",
  "-",
```

```
  as.character(original_df$Age.Group)
)
```

Check for missing values:

```
# Count num of missing values in the columns we care about
# (Some columns we will use have missing values intentionally based on schema)

col_nas <- colSums(is.na(original_df[, c(
  "Group",
  "State",
  "Sex",
  "Age.Group",
  "COVID.19.Deaths"
)]))

# Convert to a data frame with one row
col_nas_df <- as.data.frame(t(col_nas))

kable(
  col_nas_df,
  caption = "NAs per column",
  digits = 1,
  align = "c"
)
```

Table 3: NAs per column

Group	State	Sex	Age.Group	COVID.19.Deaths
0	0	0	0	39430

I have chosen not to remove the missing values (NAs) in "COVID.19.Deaths" because the CDC intentionally "suppresses" (replaces with NA) populations that are too small for confidentiality purposes. I do not want to create bias in the data by removing data from smaller populations.

Verify no "Month" values fall outside 1-12:

```r
value_range <- range(original_df$Month, na.rm = TRUE)
print(paste(
  "Range of 'Month' column values:",
  value_range[1],
  "to",
  value_range[2]
))
```

```
[1] "Range of 'Month' column values: 1 to 12"
```

Ensure no negative values for Covid-19 Deaths:

```r
min_value <- min(original_df$COVID.19.Deaths, na.rm = TRUE) # na.rm: ignore suppressed NAs
print(paste("Minimum value:", min_value))
```

```
[1] "Minimum value: 0"
```

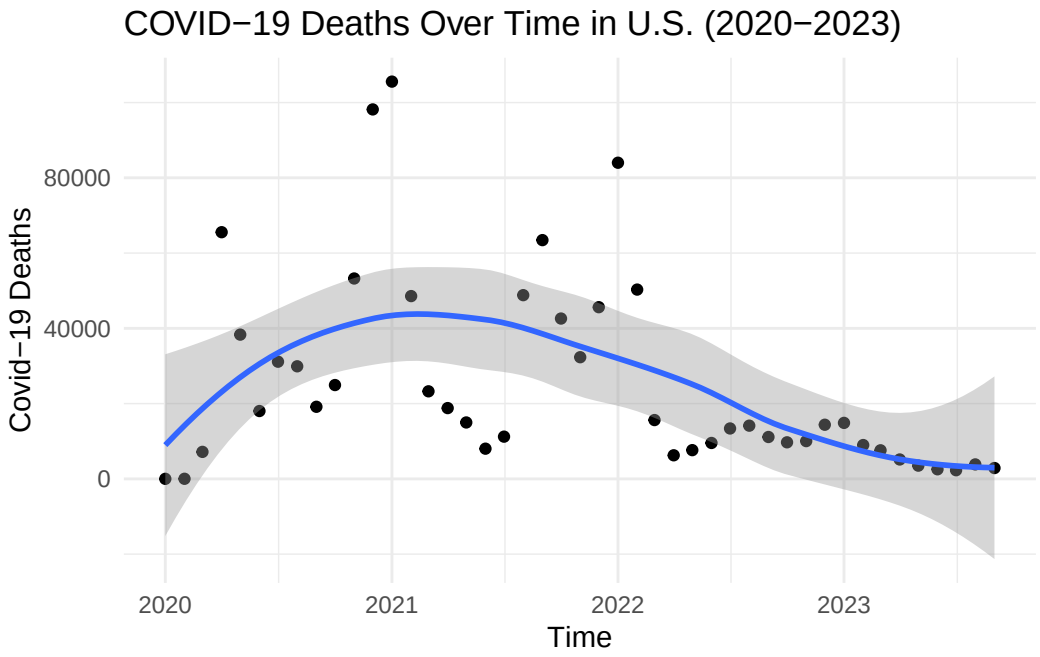
Plot total COVID-19 deaths over time in the U.S.

```r
deaths_per_month_df <- original_df %>%
  filter(
    Group == "By Month" &
      Sex == "All Sexes" &
      Age.Group == "All Ages" &
      State == "United States"
  ) |>
  dplyr::select(Year, Month, State, Sex, Age.Group, COVID.19.Deaths) %>%
  mutate(
    MonthDate = ymd(paste(Year, Month, "01", sep = "-"))
  )

# deaths_per_month_df

ggplot(data = deaths_per_month_df, aes(x = MonthDate, y = COVID.19.Deaths)) +
  geom_point() +
  geom_smooth() +
  labs(
```

```
    title = "COVID-19 Deaths Over Time in U.S. (2020-2023)",
    x = "Time",
    y = "Covid-19 Deaths"
  ) +
  theme_minimal()
```

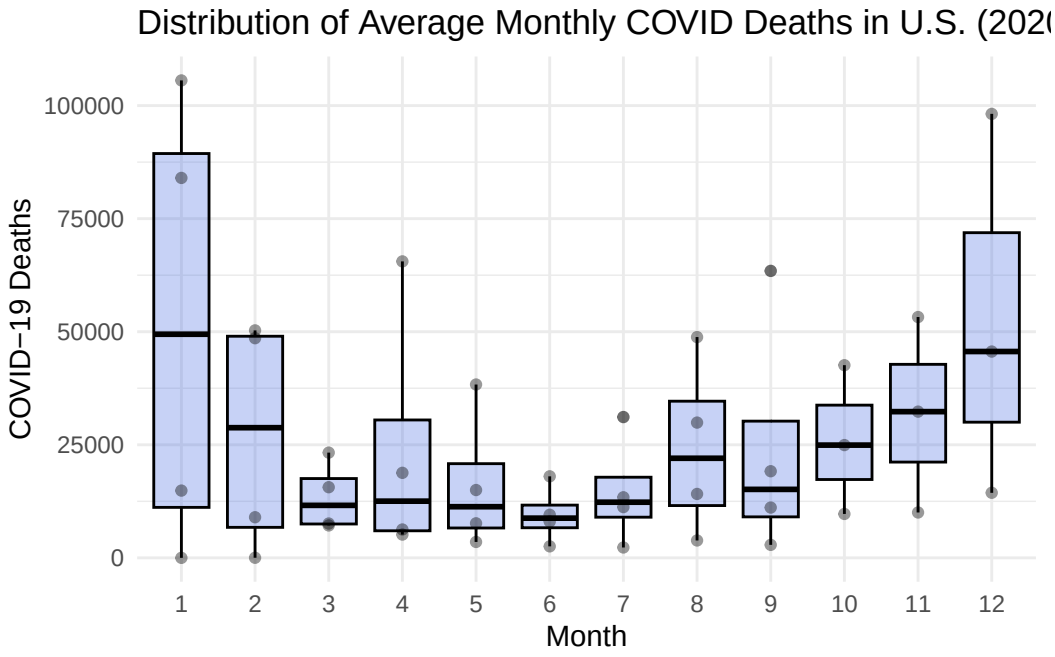


Deaths from COVID-19 spiked drastically in late 2020 and early 2021, with the death rate lowering in mid 2021 and leveling off by early 2023. By mid-2021, the US began administering booster shots, over 70% of the US had received at least one dose of the vaccine, and 60% were fully vaccinated. This could help to explain how the national death rate began declining at about the same time, since the vaccine has been an effective way to prevent covid-related hospitalizations and transmission of the virus.

Did certain months have higher average deaths across all years?

```
# Boxplot for each month across all years in the dataset
ggplot(
  data = deaths_per_month_df,
  aes(x = factor(Month), y = COVID.19.Deaths)
) +
```

```
  geom_boxplot(color = "black", fill = "royalblue", alpha = 0.3) +
  geom_point(position = position_jitter(width = 0), alpha = 0.4) +
  labs(
    title = "Distribution of Average Monthly COVID Deaths in U.S. (2020-2023)",
    x = "Month",
    y = "COVID-19 Deaths"
  ) +
  theme_minimal()
```




```
# summary(deaths_per_month_df)
```

Based on the boxplot, we can see that the average deaths per month across all years (2020-2023) peaked in January and December. This indicates a clear pattern of there being higher death rates in colder months, suggesting that the cold weather increases likelihood and spread of the virus. This may be due to the cold and dry air helping the virus survive longer and compromising people's immune systems, in addition to people spending more time indoors in close contact, allowing it to spread more easily.

How did deaths vary across age groups in the US?

```
deaths_by_ages <- original_df %>%
  filter(
    Group == "By Total" &
      Sex == "All Sexes" &
      State == "United States" &
      Age.Group != "All Ages"
  ) |>
  dplyr::select(Age.Group, COVID.19.Deaths)

# I realized there are overlapping age groups so I have to remove the redundant ones
deaths_by_ages_clean <- deaths_by_ages %>%
  filter(
    !Age.Group %in%
      c(
        "0-17 years",
        "18-29 years",
        "30-39 years",
        "40-49 years",
        "50-64 years"
      )
  )

kable(
  deaths_by_ages_clean,
  caption = "Total COVID-19 Deaths per Age group",
  digits = 1,
  col.names = c("Age Group", "Total Deaths"),
  align = "l"
)
```

Table 4: Total COVID-19 Deaths per Age group

Age Group	Total Deaths
Under 1 year	519
1-4 years	285
5-14 years	509
15-24 years	3021
25-34 years	12401
35-44 years	30108

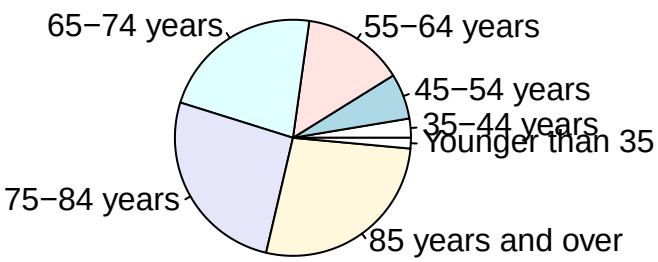
Age Group	Total Deaths
45-54 years	71388
55-64 years	159712
65-74 years	256806
75-84 years	300162
85 years and over	311863

```r
# Add threshold to group together smaller slices
threshold <- 0.02 * sum(deaths_by_ages_clean$COVID.19.Deaths)

# Group the smaller slices
deaths_by_ages_clean$group <- ifelse(
  deaths_by_ages_clean$COVID.19.Deaths < threshold,
  "Younger than 35",
  as.character(deaths_by_ages_clean$Age.Group)
)

agg <- aggregate(COVID.19.Deaths ~ group, deaths_by_ages_clean, sum)

# Plot as pie chart
pie(
  agg$COVID.19.Deaths,
  labels = agg$group,
  main = "Total Deaths by Age Group in the U.S. (2020-2023)"
)
```



Choropleth heatmap:

First, we need to get the populations of each state by doing a left join of US census data on the State column of the original dataset. Below are the first few rows of each of the datasets used in the join:

```
# Right dataset
state_pops_clean <- state_populations_data |>
  dplyr::select(
    NAME,
    POPESTIMATE2020,
    POPESTIMATE2021,
    POPESTIMATE2022,
    POPESTIMATE2023
  ) %>%
  rename(State = NAME)

head_state_pops <- head(state_pops_clean, 3)

kable(
  head_state_pops,
  caption = "State Populations Dataset (right data)",
```

```
  digits = 1,
  col.names = c(
    "State",
    "2020 Population",
    "2021 Population",
    "2022 Population",
    "2023 Population"
  ),
  align = "l"
)
```

Table 5: State Populations Dataset (right data)

State	2020 Population	2021 Population	2022 Population	2023 Population
United States	331526933	332048977	333271411	334914895
Northeast Region	57430477	57243423	57026847	56983517
New England	15057898	15106108	15120739	15159777

```
# Total state population, one row per state-year (used only for the map)
state_pops_long <- state_pops_clean %>%
  pivot_longer(
    cols = starts_with("POPESTIMATE"),
    names_to = "Year",
    values_to = "Population"
  ) %>%
  mutate(Year = as.integer(gsub("POPESTIMATE", "", Year))) # "POPESTIMATE2020" -> 2020

# 11 age groups that do not overlap and together cover all ages.
# The CDC file also has overlapping groups (e.g., "0-17 years" overlaps
# "1-4 years" and "5-14 years"); keeping both would count deaths twice.
age_groups_keep <- c(
  "Under 1 year",
  "1-4 years",
  "5-14 years",
  "15-24 years",
  "25-34 years",
  "35-44 years",
  "45-54 years",
  "55-64 years",
  "65-74 years",
  "75-84 years",
```

```
  "85 years and over"
)

left_df <- original_df %>%
  filter(
    Group == "By Year", # one row per state-year-sex-age group
    Sex != "All Sexes", # keep Male and Female only
    Age.Group %in% age_groups_keep # drops "All Ages" and the overlapping groups
  ) %>%
  mutate(
    State = as.character(State), # factor -> text so joins match on names
    Sex = as.character(Sex),
    # CDC lists New York City separately from the rest of New York State,
    # but the Census "New York" population includes NYC, so combine them
    State = if_else(State == "New York City", "New York", State),
    # parse_date_time() (lubridate) reads either "2023-09-23" or "09/23/2023"
    start = as.Date(parse_date_time(
      as.character(Start.Date),
      orders = c("ymd", "mdy")
    )),
    end = as.Date(parse_date_time(
      as.character(End.Date),
      orders = c("ymd", "mdy")
    )),
    # Share of the year these counts cover: about 1 for 2020-2022,
    # about 0.73 for 2023 (data end in September 2023)
    year_fraction = (as.numeric(end - start) + 1) / 365.25
  ) %>%
  group_by(State, Year, Sex, Age.Group, year_fraction) %>%
  # sum() WITHOUT na.rm: if NYC or rest-of-NY is suppressed (NA), the combined
  # count is unknown, so it stays NA and is excluded later by drop_na()
  summarise(COVID.19.Deaths = sum(COVID.19.Deaths), .groups = "drop")


# Expect about 1.00 for 2020-2022, about 0.73 for 2023, and no NA
distinct(left_df, Year, year_fraction)
```

```
# A tibble: 4 x 2
   Year year_fraction
  <int>         <dbl>
1  2020         1.00
2  2021         0.999
```

```
3  2022          0.999
4  2023          0.728
```

```
# Share of cells suppressed (NA) in each age group, states and DC only.
# Used to choose a reference age group that suppression barely affects.
suppression_by_age <- left_df %>%
  filter(!State %in% c("United States", "Puerto Rico")) %>% # match the rows used in the mode
  mutate(Age.Group = factor(Age.Group, levels = age_groups_keep)) %>% # youngest to oldest
  group_by(Age.Group) %>%
  summarise(
    share_suppressed = mean(is.na(COVID.19.Deaths)), # mean of TRUE/FALSE = proportion TRUE
    .groups = "drop"
  )

kable(
  suppression_by_age,
  caption = "Share of Suppressed Cells by Age Group",
  digits = 2,
  col.names = c("Age Group", "Share Suppressed"),
  align = "l"
)
```

Table 6: Share of Suppressed Cells by Age Group

Age Group	Share Suppressed
Under 1 year	0.49
1-4 years	0.34
5-14 years	0.48
15-24 years	0.54
25-34 years	0.38
35-44 years	0.30
45-54 years	0.15
55-64 years	0.07
65-74 years	0.03
75-84 years	0.00
85 years and over	0.00

```
head_left <- head(left_df, 3)

kable(
  head_left,
```

```
  caption = "Original Mortality Dataset (left data)",
  digits = 1,
  col.names = c(
    "State",
    "Year",
    "Sex",
    "Age Group",
    "Year Fraction",
    "COVID-19 Deaths"
  ),
  align = "l"
)
```

Table 7: Original Mortality Dataset (left data)

State	Year	Sex	Age Group	Year Fraction	COVID-19 Deaths
Alabama	2020	Female	1-4 years	1	0
Alabama	2020	Female	15-24 years	1	NA
Alabama	2020	Female	25-34 years	1	17

```
# Reshape the Census file to one row per state-year-sex-age group,
# binned into the same 11 age groups as the CDC data
age_sex_pops_long <- age_sex_pops_df %>%
  filter(
    as.numeric(SUMLEV) == 40, # state rows only ("040"; 010 = nation)
    SEX %in% c(1, 2), # 1 = Male, 2 = Female (0 = both combined)
    AGE != 999 # 999 = all ages combined
  ) %>%
  mutate(
    Sex = if_else(SEX == 1, "Male", "Female"),
    # cut() bins ages into intervals closed on the right:
    # (-Inf,0] = age 0, (0,4] = 1-4, (4,14] = 5-14, ..., (84,Inf] = 85+
    Age.Group = as.character(cut(
      AGE,
      breaks = c(-Inf, 0, 4, 14, 24, 34, 44, 54, 64, 74, 84, Inf),
      labels = age_groups_keep
    ))
  ) %>%
  dplyr::select(
    State = NAME,
    Sex,
```

```
    Age.Group,
    POPEST2020_CIV,
    POPEST2021_CIV,
    POPEST2022_CIV,
    POPEST2023_CIV
  ) %>%
  pivot_longer(
    cols = starts_with("POPEST"),
    names_to = "Year",
    values_to = "Population"
  ) %>%
  mutate(Year = as.integer(gsub("POPEST|_CIV", "", Year))) %>% # "POPEST2020_CIV" -> 2020
  group_by(State, Year, Sex, Age.Group) %>%
  summarise(Population = sum(Population), .groups = "drop") # add single ages within each bi

# Sanity check on the age binning: summed civilian age-sex populations should be
# just under each state's total resident population (civilians exclude active-duty military)
pop_check <- age_sex_pops_long %>%
  group_by(State, Year) %>%
  summarise(civilian = sum(Population), .groups = "drop") %>%
  inner_join(
    state_pops_long %>% mutate(State = as.character(State)),
    by = c("State", "Year")
  ) %>%
  mutate(ratio = civilian / Population)

summary(pop_check$ratio) # expect values just under 1
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.9692  0.9959  0.9982  0.9972  1.0001  1.0114
```

```
pop_check %>% arrange(ratio) %>% head(5) # lowest ratios: likely states with large military p
```

```
# A tibble: 5 x 5
  State   Year civilian Population ratio
  <chr>  <int>    <int>      <int> <dbl>
1 Alaska  2021   712293     734923 0.969
2 Hawaii  2022  1396594    1439399 0.970
3 Alaska  2022   711502     733276 0.970
4 Hawaii  2021  1404106    1446745 0.971
5 Alaska  2020   711560     732964 0.971
```

```r
# Join the two datasets using a left join
# Each row's Population is now that age-sex group's population in that state-year
joined_df <- left_df %>%
  left_join(age_sex_pops_long, by = c("State", "Year", "Sex", "Age.Group")) %>%
  mutate(
    # Person-years at risk: population scaled by the share of the year covered,
    # so partial-year 2023 is not compared against a full year of population
    person_years = Population * year_fraction,
    deaths_per_100000 = (COVID.19.Deaths / person_years) * 100000, # per 100,000 person-years
    State = tolower(State) # lowercase names for the map
  )

# Check: only non-states (e.g., "united states", "puerto rico") should be listed
joined_df %>% filter(is.na(Population)) %>% distinct(State)
```

```
# A tibble: 2 x 1
  State
  <chr>
1 puerto rico
2 united states
```

```r
head_joined <- head(joined_df, 3)

kable(
  head_joined,
  caption = "Left Join of Mortality and Populations Datasets",
  digits = 1,
  col.names = c(
    "State",
    "Year",
    "Sex",
    "Age Group",
    "Year Fraction",
    "COVID-19 Deaths",
    "Population",
    "Person-Years",
    "Deaths per 100,000"
  ),
  align = "l"
)
```

Table 8: Left Join of Mortality and Populations Datasets

State	Year	Sex	Age Group	Year Fraction	COVID-19 Deaths	Population	Person-Years	Deaths per 100,000
alabama	2020	Female	1-4 years	1	0	114841	115076.8	0.0
alabama	2020	Female	15-24 years	1	NA	347569	348282.7	NA
alabama	2020	Female	25-34 years	1	17	320104	320761.3	5.3

Add a "region" and death rate columns to the new joined data.

```
# Define the states in each region:
west <- c(
  "alaska",
  "hawaii",
  "washington",
  "oregon",
  "california",
  "nevada",
  "idaho",
  "montana",
  "wyoming",
  "utah",
  "colorado",
  "arizona",
  "new mexico"
)

midwest <- c(
  "north dakota",
  "south dakota",
  "nebraska",
  "kansas",
  "minnesota",
  "iowa",
  "missouri",
  "wisconsin",
  "illinois",
  "indiana",
```

```
  "michigan",
  "ohio"
)

south <- c(
  "delaware",
  "maryland",
  "virginia",
  "west virginia",
  "kentucky",
  "tennessee",
  "north carolina",
  "south carolina",
  "georgia",
  "florida",
  "alabama",
  "mississippi",
  "arkansas",
  "louisiana",
  "texas",
  "oklahoma",
  "district of columbia"
)

northeast <- c(
  "maine",
  "new hampshire",
  "vermont",
  "massachusetts",
  "rhode island",
  "connecticut",
  "new york",
  "new jersey",
  "pennsylvania"
)

# Add region column, drop territories (region = NA) and CDC-suppressed counts
# (COVID.19.Deaths = NA), and sort by year ascending
joined_regional_df <- joined_df %>%
  mutate(
    region = case_when(
      State %in% west ~ "west",
```

```
      State %in% midwest ~ "midwest",
      State %in% south ~ "south",
      State %in% northeast ~ "northeast",
      TRUE ~ NA_character_
    )
  ) %>%
  mutate(region = as.factor(region)) %>%
  drop_na() %>%
  arrange(Year)

# head(joined_regional_df, 3)
```

Prepare subsets of the joined data by each year.

```
joined_2020 <- joined_regional_df %>%
  filter(Year == "2020")

joined_2021 <- joined_regional_df %>%
  filter(Year == "2021")

joined_2022 <- joined_regional_df %>%
  filter(Year == "2022")

joined_2023 <- joined_regional_df %>%
  filter(Year == "2023")
```

Finally, plot the choropleth maps for 2020-2023:

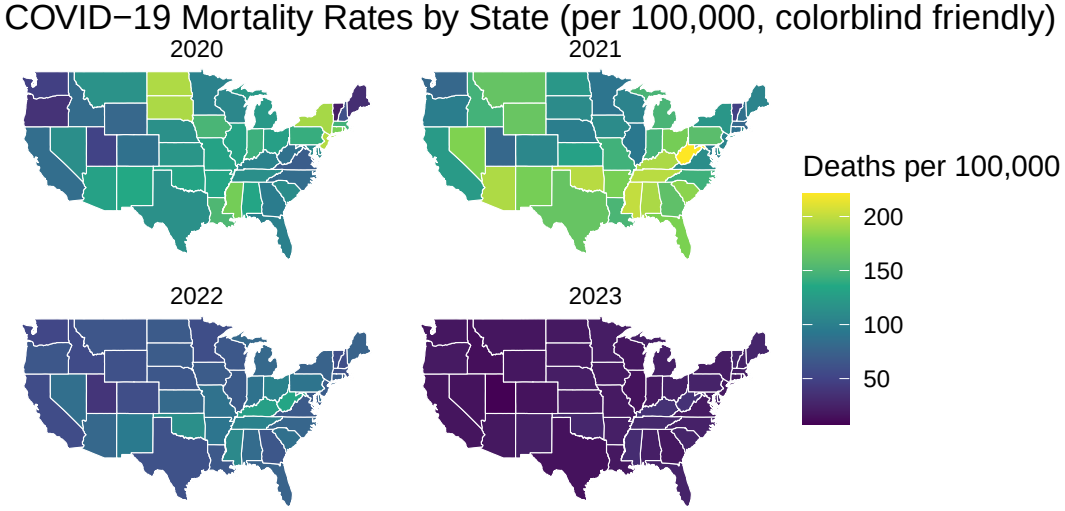
```
# Load US map geometry
us_map <- map_data("state")

# One row per state-year (all ages, both sexes) so each state gets one color.
map_df <- original_df %>%
  filter(Group == "By Year", Sex == "All Sexes", Age.Group == "All Ages") %>%
  mutate(
    State = as.character(State),
    State = if_else(State == "New York City", "New York", State),
    start = as.Date(parse_date_time(
      as.character(Start.Date),
      orders = c("ymd", "mdy")
```

```
    )),
    end = as.Date(parse_date_time(
      as.character(End.Date),
      orders = c("ymd", "mdy")
    )),
    year_fraction = (as.numeric(end - start) + 1) / 365.25
  ) %>%
  group_by(State, Year, year_fraction) %>%
  summarise(COVID.19.Deaths = sum(COVID.19.Deaths), .groups = "drop") %>%
  left_join(
    state_pops_long %>% mutate(State = as.character(State)), # total state population
    by = c("State", "Year")
  ) %>%
  mutate(
    deaths_per_100000 = COVID.19.Deaths / (Population * year_fraction) * 100000,
    State = tolower(State)
  )

plot_map_df <- us_map %>%
  left_join(map_df, by = c("region" = "State"))

# Plot the heatmaps on a 2x2 grid by year
ggplot(plot_map_df, aes(long, lat, group = group, fill = deaths_per_100000)) +
  geom_polygon(color = "white", linewidth = 0.2) +
  coord_fixed(1.3) +
  scale_fill_viridis_c(option = "viridis") +
  facet_wrap(~Year, ncol = 2) +
  labs(
    title = "COVID-19 Mortality Rates by State (per 100,000, colorblind friendly)",
    fill = "Deaths per 100,000"
  ) +
  theme_minimal() +
  theme_void() + # Remove background and axes
  theme(
    axis.title = element_blank(),
    axis.text = element_blank(),
    axis.ticks = element_blank()
  )
```



Statistical Analysis:

Can we predict death rates using the year, region and sex variables?

Multiple Linear Regression:

```
# monthly_age_sex_deaths <- original_df %>%
#   filter(Group == "By Month") %>%
#   distinct(Year, Month, State, Sex, Age.Group, .keep_all = TRUE) %>%
#   select(Group, Year, Month, State, Sex, Age.Group, COVID.19.Deaths)

# # str(monthly_age_sex_deaths)

# summary(joined_regional_df)

joined_regional_no_AllAges_df <- joined_regional_df %>%
  mutate(
    # Youngest to oldest, so tables and plots read in order
```

```r
    Age.Group = factor(Age.Group, levels = age_groups_keep),
    # Reference group that all other age groups are compared to
    Age.Group = relevel(Age.Group, ref = "5-14 years"),
    Sex = factor(Sex, levels = c("Female", "Male")) # Female = baseline, as before
  )
# summary(joined_regional_no_AllAges_df)

mlr_model <- lm(
  deaths_per_100000 ~ factor(Year) + Age.Group + region + Sex,
  data = joined_regional_no_AllAges_df
)

summary(mlr_model)
```

```
Call:
lm(formula = deaths_per_100000 ~ factor(Year) + Age.Group + region +
    Sex, data = joined_regional_no_AllAges_df)

Residuals:
    Min      1Q  Median      3Q     Max
-1114.6  -101.2   -20.0    90.8  3675.3

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                  98.524     21.912   4.496 7.15e-06 ***
factor(Year)2021            -25.698     12.638  -2.033  0.04209 *
factor(Year)2022           -138.933     13.041 -10.653  < 2e-16 ***
factor(Year)2023           -299.724     13.170 -22.759  < 2e-16 ***
Age.GroupUnder 1 year         1.448     25.901   0.056  0.95542
Age.Group1-4 years           -8.744     24.440  -0.358  0.72054
Age.Group15-24 years          4.464     26.718   0.167  0.86731
Age.Group25-34 years        -37.090     24.977  -1.485  0.13765
Age.Group35-44 years        -30.469     24.298  -1.254  0.20994
Age.Group45-54 years         14.351     23.315   0.616  0.53824
Age.Group55-64 years         72.128     22.904   3.149  0.00165 **
Age.Group65-74 years        173.134     22.704   7.626 3.14e-14 ***
Age.Group75-84 years        441.412     22.596  19.535  < 2e-16 ***
Age.Group85 years and over 1288.131     22.586  57.032  < 2e-16 ***
regionnortheast             -27.157     14.542  -1.867  0.06193 .
regionsouth                   8.431     12.419   0.679  0.49729
regionwest                  -56.325     13.126  -4.291 1.83e-05 ***
```

```
SexMale                       92.528      9.204  10.053  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 266.3 on 3334 degrees of freedom
Multiple R-squared:  0.7269,    Adjusted R-squared:  0.7255
F-statistic:   522 on 17 and 3334 DF,  p-value: < 2.2e-16
```

Comments:

- The intercept is: Year = 0 (redundant), region = midwest, and Sex = Female

- We converted the Year variable into a factor for the MLR here because COVID death rates do not decline linearly over time. Treating the years as categorical allows us to fit separate means for each year, captures non-linear year effects, and reduces the residual patterns

Interpretation:

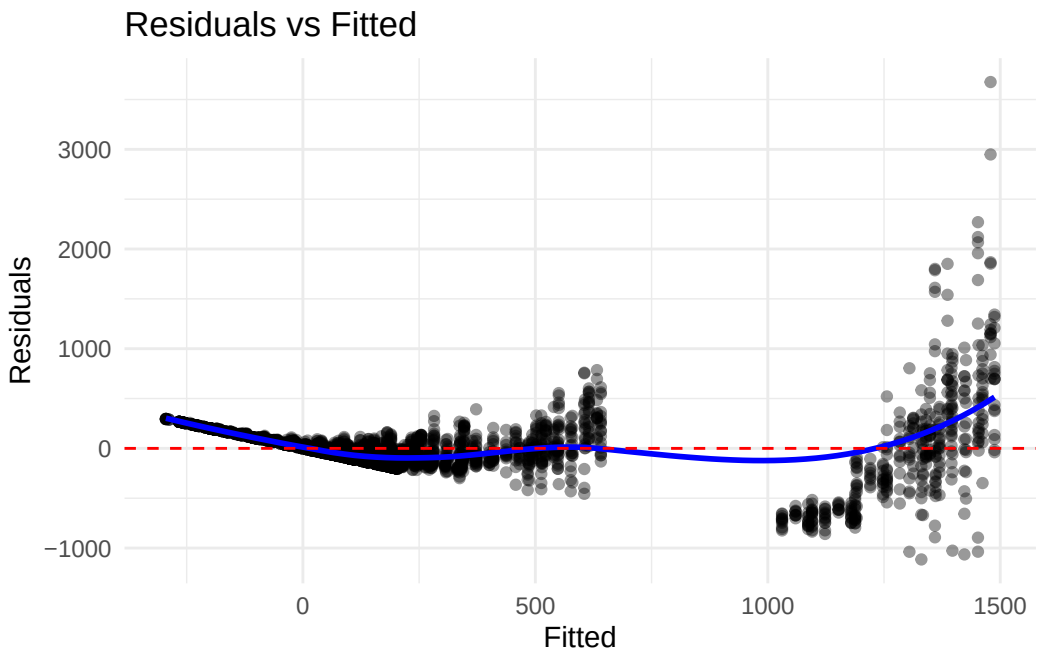
- The model explains about 63.6% of the variation in COVID-19 deaths per 100,000
  - Adjusted $R^2 = 0.636$
- COVID mortality peaked in 2021, then sharply declined in 2022–2023
  - Mortality rates were 1.63 per 100k higher than 2020
- All age groups below 40 years old had no statistically significant difference from the baseline age group of 0-17 years
- The statistical sifgnificance of age group increases as age groups increase, with the 85+ age group exhibiting the highest mortality risk relative to the baseline
- The south had significantly higher COVID mortality per 100k than all other regions
  - States in the South are expected to have COVID-19 death rates that are about 0.66 deaths per 100,000 population higher than the Midwest, after accounting for differences in year, age group, and sex

Diagnostic plots for regression:

```r
library(ggplot2)
library(broom)

# Residuals vs Fitted
# Check linearity and homoscedasticity
ggplot(mlr_model, aes(.fitted, .resid)) +
```

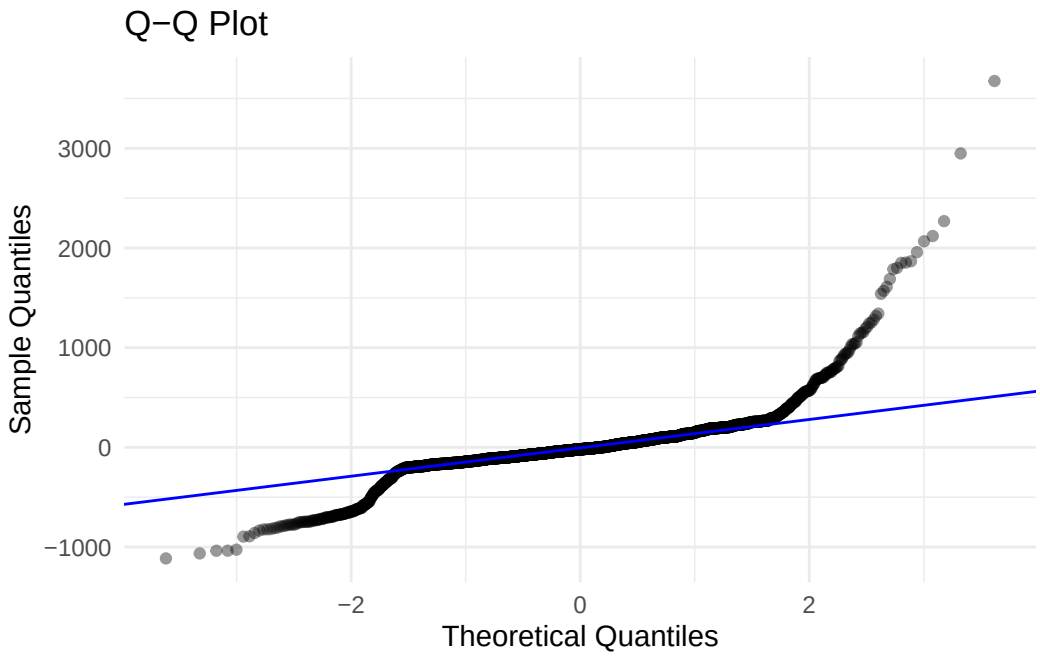
```
  geom_point(alpha = 0.4) +
  geom_smooth(method = "loess", se = FALSE, color = "blue") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Residuals vs Fitted", x = "Fitted", y = "Residuals") +
  theme_minimal()
```



The Residuals vs Fitted plot demonstrates that the model clearly violates the asumption of linearity. By looking at the plot, we can see that the predictors (year, region, and sex) and the response variable (death rate) are not linearly related. The wavy LOESS curve demonstrates an uneven spread around the residuals = 0 line, indicating that the model is potentially overpredicting in some ranges and underpredicting in others. The assumption of constant variance (homoscedasticity) is also violated, exhibited by the widening funnel shape in the plot. The residuals at lower fitted values are closer together, while those at higher fitted values spread out more dramatically. This means that the variance in error grows as the predicted mortality increases.

```
# Q-Q Plot
# Check normality of residuals
ggplot(mlr_model, aes(sample = .resid)) +
  stat_qq(alpha = 0.4) +
  stat_qq_line(color = "blue") +
```

```
  labs(
    title = "Q-Q Plot",
    x = "Theoretical Quantiles",
    y = "Sample Quantiles"
  ) +
  theme_minimal()
```



The Q-Q plot demonstrates that the model violates the asumption of normality of the residuals. The points slightly deviate below the line at and below theoretical quantiles of -3, while deviating much further above the line as the quantiles increase in value. The data is light-tailed, suggesting the data may be right-skewed and the model potentially underestimating high death-rate observations.

Poisson Regression:

Use Poission regression to more accurately model the COVID deaths, since our data is count data not continuous.

```
poisson_regression <- glm(
  COVID.19.Deaths ~ Sex + region + factor(Year) + Age.Group, # Year as factor, as in the MLR
  family = poisson,
```

```r
  offset = log(person_years), # group-specific person-years at risk
  data = joined_regional_no_AllAges_df
)

summary(poisson_regression)
```

```
Call:
glm(formula = COVID.19.Deaths ~ Sex + region + factor(Year) +
    Age.Group, family = poisson, data = joined_regional_no_AllAges_df,
    offset = log(person_years))

Coefficients:
                            Estimate Std. Error  z value Pr(>|z|)
(Intercept)               -13.599900   0.124058 -109.626   <2e-16 ***
SexMale                     0.453870   0.001889  240.232   <2e-16 ***
regionnortheast             0.027882   0.002972    9.381   <2e-16 ***
regionsouth                 0.098074   0.002508   39.101   <2e-16 ***
regionwest                 -0.148278   0.002948  -50.292   <2e-16 ***
factor(Year)2021            0.168406   0.002182   77.184   <2e-16 ***
factor(Year)2022           -0.487052   0.002585 -188.444   <2e-16 ***
factor(Year)2023           -1.736030   0.004734 -366.711   <2e-16 ***
Age.GroupUnder 1 year       2.135869   0.190310   11.223   <2e-16 ***
Age.Group1-4 years         -0.321938   0.238841   -1.348    0.178
Age.Group15-24 years        2.853819   0.125892   22.669   <2e-16 ***
Age.Group25-34 years        4.136522   0.124377   33.258   <2e-16 ***
Age.Group35-44 years        5.042652   0.124172   40.610   <2e-16 ***
Age.Group45-54 years        5.940017   0.124092   47.868   <2e-16 ***
Age.Group55-64 years        6.678914   0.124060   53.836   <2e-16 ***
Age.Group65-74 years        7.387841   0.124051   59.555   <2e-16 ***
Age.Group75-84 years        8.249663   0.124049   66.504   <2e-16 ***
Age.Group85 years and over  9.308711   0.124048   75.041   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 2719122  on 3351  degrees of freedom
Residual deviance:  135113  on 3334  degrees of freedom
AIC: 152342

Number of Fisher Scoring iterations: 6
```

Interpretation:

- The COVID-19 mortality rate for males is around 28% higher than for females
  - IRR = exp(0.2448) 1.28
- The Northeast had a 14% lower COVID-19 mortality rate than in the Midwest
  - IRR = exp(−0.1501) 0.86
- The South had a 10% higher COVID-19 mortality rate than the Midwest
  - IRR = exp(0.0938) 1.10
- The West had a 16% lower mortality rate than the Midwest
  - IRR = exp(−0.1707) 0.84
- Each additional year predicts a 35% decrease in the COVID-19 mortality rate
  - IRR = exp(−0.4322) 0.65
- Older adults showed much higher COVID-19 mortality rates, with the 85+ group having a more than 150 times higher death rate than 0-17 year olds
- Overall, COVID-19 mortality rates grow with age, which is the strongest predictor of death in our Poisson regression model

Negative Binomial Regression:

```
library(MASS)

# Define model, using population as offset to scale porportionally
neg_binomial_regression <- glm.nb(
  COVID.19.Deaths ~ Sex +
    region +
    factor(Year) +
    Age.Group +
    offset(log(person_years)),
  data = joined_regional_no_AllAges_df
)

summary(neg_binomial_regression)
```

```
Call:
```

```
glm.nb(formula = COVID.19.Deaths ~ Sex + region + factor(Year) +
    Age.Group + offset(log(person_years)), data = joined_regional_no_AllAges_df,
    init.theta = 5.641447023, link = log)

Coefficients:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)               -14.00703    0.15295 -91.582  < 2e-16 ***
SexMale                     0.43071    0.01735  24.823  < 2e-16 ***
regionnortheast            -0.14615    0.02789  -5.241  1.6e-07 ***
regionsouth                 0.24149    0.02291  10.539  < 2e-16 ***
regionwest                 -0.04334    0.02505  -1.730   0.0836 .
factor(Year)2021            0.47148    0.02319  20.333  < 2e-16 ***
factor(Year)2022           -0.31238    0.02376 -13.146  < 2e-16 ***
factor(Year)2023           -1.71406    0.02710 -63.250  < 2e-16 ***
Age.GroupUnder 1 year       2.21883    0.22411   9.901  < 2e-16 ***
Age.Group1-4 years         -0.15488    0.26272  -0.590   0.5555
Age.Group15-24 years        2.90979    0.15892  18.310  < 2e-16 ***
Age.Group25-34 years        4.27999    0.15447  27.708  < 2e-16 ***
Age.Group35-44 years        5.16062    0.15374  33.567  < 2e-16 ***
Age.Group45-54 years        6.02764    0.15324  39.335  < 2e-16 ***
Age.Group55-64 years        6.78858    0.15302  44.364  < 2e-16 ***
Age.Group65-74 years        7.56033    0.15294  49.433  < 2e-16 ***
Age.Group75-84 years        8.53536    0.15291  55.819  < 2e-16 ***
Age.Group85 years and over  9.70066    0.15291  63.442  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(5.6414) family taken to be 1)

    Null deviance: 68662.0  on 3351  degrees of freedom
Residual deviance:  3220.9  on 3334  degrees of freedom
AIC: 28997

Number of Fisher Scoring iterations: 1

              Theta:  5.641
          Std. Err.:  0.174

 2 x log-likelihood:  -28958.667
```

```
neg_binomial_regression$converged # should be TRUE; FALSE means the estimates can't be truste
```

```
[1] TRUE
```

```
neg_binomial_regression$theta # estimated dispersion parameter k
```

```
[1] 5.641447
```

Interpretation:

- Male COVID-19 mortality rates are about 37% higher than female mortality rates
  - IRR = exp(0.31441) 1.37
- The northeast had ~24% lower mortality rates than the midwest
  - IRR = exp(−0.27157) 0.76
- The south had ~36% higher mortality rates than the midwest
  - exp(0.30610) 1.36
- The west had ~4% higher mortality rates than the midwest
  - exp(-1.707e-01) 0.84
- Each additional year predicts a 44% decrease in mortality rates
  - exp(−0.58368) 0.56
- The 85+ year old group had a more than 190 times higher death rate than 0-17 year olds
- Similarly to the Poisson regression model, the Negative Binomial regression model predicts that COVID-19 mortality rates increase with age

Compare Poisson vs Negative Binomial Regression:

```
aic <- AIC(poisson_regression, neg_binomial_regression)
kable(aic, caption = "AIC Results", digits = 1, align = "lcr")
```

Table 9: AIC Results

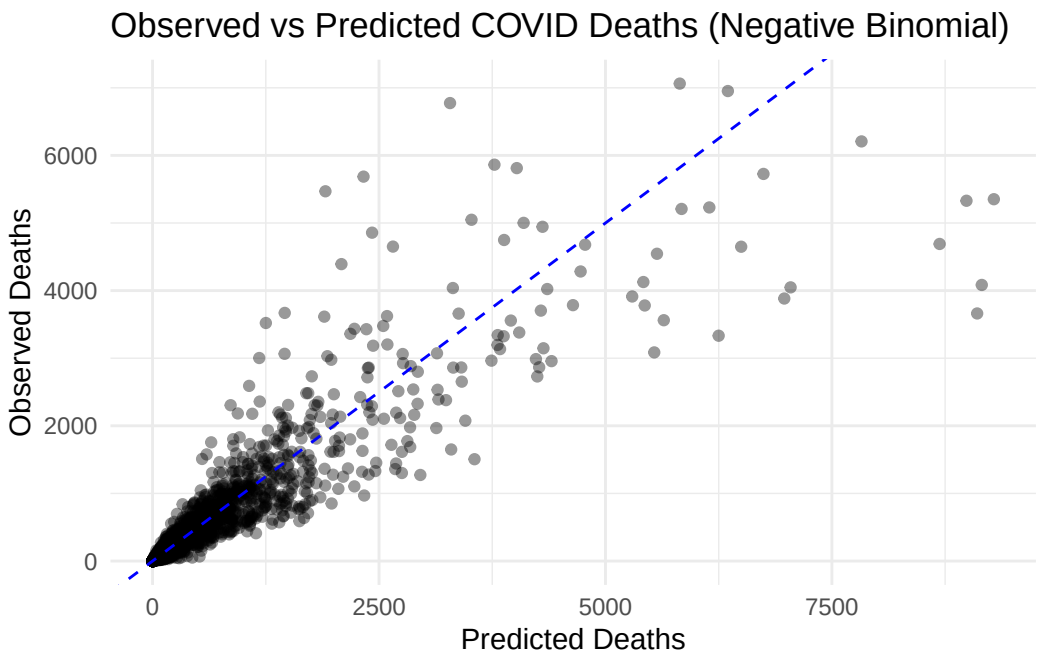
	df	AIC
poisson_regression	18	152341.9
neg_binomial_regression	19	28996.7

The AIC (Akaike Information Criterion) comparison table comfirms that the Negative Binomial regression model performs better than poisson regression in predicting COVID mortality rates. The negative binomial regression model has a much lower AIC value of 44,747.3 compared to that of the Poisson Regression model (480,266.0). This demonstrates that the Negative Binomial approach exhibits a lower estimated prediction error and therefore better fits the mortality data.

```
# exp(cbind(
#   IRR = coef(neg_binomial_regression),
#   confint(neg_binomial_regression)
# ))

joined_regional_no_AllAges_df$pred_nb <- predict(
  neg_binomial_regression,
  type = "response"
)

ggplot(joined_regional_no_AllAges_df, aes(x = pred_nb, y = COVID.19.Deaths)) +
  geom_point(alpha = 0.4) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "blue") +
  labs(
    title = "Observed vs Predicted COVID Deaths (Negative Binomial)",
    x = "Predicted Deaths",
    y = "Observed Deaths"
  ) +
  theme_minimal()
```



The Observed vs Predicted COVID Deaths plot shows that the Negative Binomial Regression model is both underpredicting and overpredicting covid deaths. Notably, the fan-shaped pattern indicates that the model's prediction accuracy lessens as the number of deaths increase. Despite this, the model does a decent job of predicting the number of COVID deaths when the observed deaths are below about 60,000.

Age × Sex Interaction (Likelihood Ratio Test):

Does the effect of age on COVID-19 mortality rates differ by sex?

```
# Same NB model, plus the age x sex interaction
nb_interaction <- glm.nb(
  COVID.19.Deaths ~ Sex *
    Age.Group +
    region +
    factor(Year) +
    offset(log(person_years)),
  data = joined_regional_no_AllAges_df
)
```

```r
nb_interaction$converged # should be TRUE, same check as the main model
```

```
[1] TRUE
```

```r
# Likelihood ratio test: does adding the interaction significantly improve fit?
anova(neg_binomial_regression, nb_interaction)
```

```
Likelihood ratio tests of Negative Binomial Models

Response: COVID.19.Deaths
                                                                        Model    theta
1 Sex + region + factor(Year) + Age.Group + offset(log(person_years)) 5.641447
2 Sex * Age.Group + region + factor(Year) + offset(log(person_years)) 5.669274
  Resid. df    2 x log-lik.   Test    df LR stat.    Pr(Chi)
1      3334       -28958.67
2      3324       -28938.11 1 vs 2    10   20.558 0.02439541
```

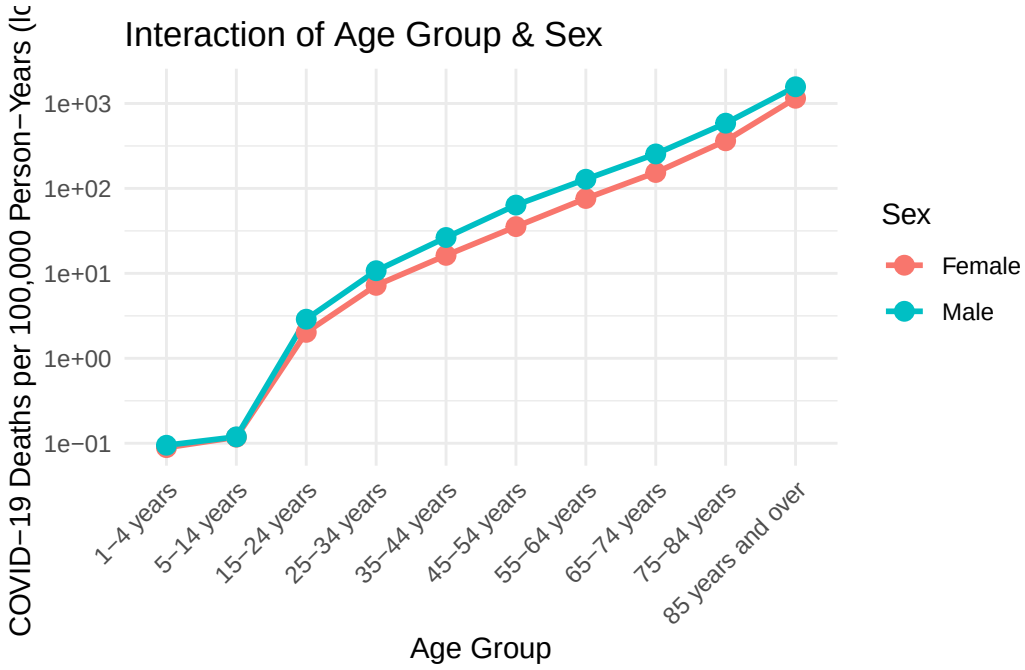
Interpretation:

- Age group, sex, and region eachhave statistically significant effects on COVID-19 mortality rates in the US, as they have p-values less than the alpha level of 0.05. This means that, mortality rates differ across age groups, between sexes, across regions when averaged over the other factors.
- The interaction between age group and sex is statistically significant as well since it also has a p-value less than the alpha level of 0.05. This means that the effect of age group on mortality rates depends on sex (and changes as sex changes), and vice versa. In the data this indicates that the difference in mortality rates between age groups is not consistent across sexes. Similarly, the difference in mortality rates between sexes is not consistent across age groups.
- The interactions between Age Group x Region, Sex x Region, & Age Group x Sex x Region are not statistically significant. This means there is no evidence that the combined effect of those pairs (or all three together) influences mortality outside of what is already explained by the individual main effects and the interaction between Age Group $\times$ Sex.

Interaction plot:

```r
interaction_subset <- joined_regional_no_AllAges_df %>%
  filter(Age.Group != "Under 1 year") %>%
  mutate(Age.Group = factor(Age.Group, levels = age_groups_keep)) %>%
  group_by(Age.Group, Sex) %>%
  summarise(
    # pooled rate: total deaths / total person-years, so large and small states are weighted
    rate = sum(COVID.19.Deaths) / sum(person_years) * 100000,
    .groups = "drop"
  )
# In the ggplot: aes(y = rate), and add + scale_y_log10()
# (on a log scale, parallel lines = no age x sex interaction in a multiplicative model)

# Create the interaction plot (rates, not counts)
ggplot(
  interaction_subset,
  aes(x = Age.Group, y = rate, group = Sex, color = Sex) # was y = mean_deaths
) +
  geom_line(linewidth = 1) + # "size" is deprecated for lines in current ggplot2
  geom_point(size = 3) +
  scale_y_log10() + # log scale: parallel lines = no age x sex interaction
  labs(
    title = "Interaction of Age Group & Sex",
    x = "Age Group",
    y = "COVID-19 Deaths per 100,000 Person-Years (log scale)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
# summary(joined_regional_no_AllAges_df)
```

Interpretation:

- From ages 0-44 years old, the interaction plot suggests that sex has little effect on predicted COVID-19 deaths
- According to the plot, being male increases COVID-19 deaths much more in those aged 45-84 years old than being female, as the average COVID-19 deaths for males is much higher in that age bracket
- After age 84, the effect of sex on predicted COVID-19 deaths reverses depending on age group, indicating an interaction between the two predictive variables

---

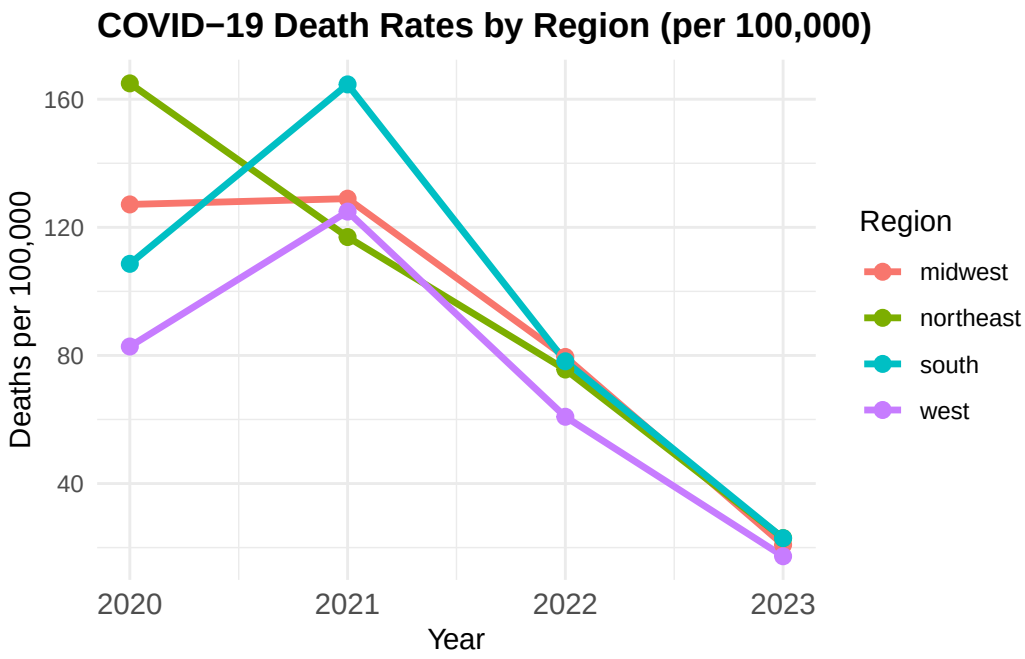
Panel data plot:

```
# summary(joined_regional_no_AllAges_df)

region_year_panel <- map_df %>%
  mutate(
```

```
    region = case_when(
      State %in% west ~ "west",
      State %in% midwest ~ "midwest",
      State %in% south ~ "south",
      State %in% northeast ~ "northeast"
    )
  ) %>%
  filter(!is.na(region), !is.na(COVID.19.Deaths)) %>%
  group_by(region, Year) %>%
  summarise(
    deaths = sum(COVID.19.Deaths),
    person_years = sum(Population * year_fraction), # total state population, adjusted for pa
    deaths_per_100000 = deaths / person_years * 100000,
    .groups = "drop"
  )

ggplot(
  region_year_panel,
  aes(x = Year, y = deaths_per_100000, color = region, group = region)
) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2.5) + # optional-remove this line if you don't want points
  labs(
    title = "COVID-19 Death Rates by Region (per 100,000)",
    x = "Year",
    y = "Deaths per 100,000",
    color = "Region"
  ) +
  scale_x_continuous(
    breaks = c(2020, 2021, 2022, 2023),
    labels = c("2020", "2021", "2022", "2023")
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold"),
    axis.text.x = element_text(size = 11)
  )
```



Interpretation:

- All regions show a clear decline in COVID-19 mortality from 2021 to 2023
- Mortality peaked in 2021 for every region
- The South has the highest mortality peak, while the Northeast shows the lowest overall rates
- By 2023, mortality drops to very low levels across all regions, indicating effective vaccinations, improved treatment, and reduced severity in later variants